UNIVERSITY OF HYDERABAD
SCHOOL OF ENGINEERING SCIENCES & TECHNOLOGY
CENTRAL UNIVERSITY P.O, PROF. C. R. RAO ROAD, GACHIBOWLI,
HYDERABAD –500046, INDIA

22/12/2025

Ref. No.: UoH/SEST/VGK/DST-AMT/HiPIMS

Global tender enquiry for the supply and commissioning of HiPIMS power supply

The School of Engineering Sciences & Technology (SEST), University of Hyderabad (UoH), invites tenders in a two-bid format from highly reputed and established firms and authorised dealers for the: Supply, installation, testing and commissioning of Bipolar HiPIMS power supply capable of handling reactive sputtering process. **Final Terms and Conditions are as per UoH's latest purchase policy.** The tenders must reach the below given address on or before 5.00 PM on **January 19, 2026.**

THE FOLLOWING CONDITIONS MUST BE STRICTLY MET IF THE QUOTATION IS TO BE VALID

1. All quotations should be addressed to, and envelopes should be superscribed with the following: " HiPIMS power supply":

Dr. Venkata Girish Kotnur
School of Engineering Sciences & Technology,
West Campus, University of Hyderabad
Central University P.O., Gachibowli,
Hyderabad 500046, INDIA.

2. Quotations must reach before 5 PM on 19.01.2026. Quotations received after the due date and time will not be considered. The University of Hyderabad will not, in any way, be responsible for any postal or other delays.

3. The bids shall be submitted in a TWO-BID format in separate sealed covers. Item code, enquiry number & due date must be compulsorily mentioned on both envelopes contained within another main envelope duly sealed, which shall also have item code, enquiry number + due date superscribed.

a) **TECHNO-COMMERCIAL BID** – Please give detailed specifications, International standards (BIS / INTERNATIONAL), catalogues, list of users & technical details, technical compliance sheet, etc., along with the operating parameters, pre-installation requirements, payment terms, warranty/guarantee period, delivery period, etc., very explicitly without any ambiguity. The quotation must clearly specify the make and model of the equipment. All relevant technical literature/brochures, application notes, and specifications must also be provided, adequately explaining and confirming compliance of the features of the model of the equipment being quoted with those given in the UoH tender enquiry. Without this, the quotation may be considered invalid. It must be clearly indicated if any of the features do not match. Explanations on the non-matching specifications will be considered. The University of Hyderabad reserves the right to determine which deviations are substantial based on current and future

requirements. If features in the quoted equipment are better than those indicated in the enquiry, they must also be clearly explained.

b) **PRICE BID** - giving full prices in foreign currency or in Indian rupees for, Main equipment, Essential accessories & spares, All optional items, Annual Maintenance Contract (AMC) charges for three years after the expiry of the warranty period.

4. Price & insurance:

The price quoted should be as follows:

In case of Indian suppliers :Insurance and freight prepaid by the vendor, Free delivery at the School of Engineering Sciences and Technology, University of Hyderabad.

In case of foreign suppliers/import:

CIP, HYDERABAD AIRPORT BASIS

Note: The offers of the firms that quoted on Ex-works basis and do not mention the handling, documentation, packing, forwarding, transportation & insurance charges, etc., separately will not be considered as a responsive bid, and the acceptance of such bid will vest entirely with the University of Hyderabad

5. Vendor qualification criteria for acceptance of technical Bid :

a) Vendor should have supplied at least 3 HiPIMS system to Govt Organizations/Research Institutes in the past 5 years. Copy of Installation Report / Certificate must be submitted.

b) Vendor must explicitly mention and provide proof of the items claimed under Make in India Scheme.

c) Vendor must visit University of Hyderabad with their HiPIMS System and demonstrate compatibility, stability, and operational capability of their HiPIMS technology. All travel, logistics, transport, and demonstration costs shall be borne entirely by the Vendor. No financial responsibility shall be placed on the University for these demonstrations.

ALL OPTIONS SHOULD BE QUOTED SEPARATELY.

6. Validity of prices:

The quoted prices must be valid for a minimum period of 180 days from the date of opening of the financial bids, failing which offers may be summarily rejected by the University of Hyderabad.

7. Compliance statement:

a) All manufacturers/agents must submit the compliance statement and the technical bid.

b) Please provide the following details if you have supplied similar or identical items in India, particularly to other research/academic institutions in **the last three years.**

- **Name of the item with brief specifications, model etc.,**
- **Name & contact address of the user.**
- **Any other relevant information**.

8. Installation & commissioning:

Installation & commissioning will be the sole responsibility of the supplier / Indian agent. The supplier must do installation & commissioning with all infrastructural works. All vendors/agents must submit full details and requirements for installation & commissioning of the items as per the technical specifications they submitted.

9. Taxes, duties, and levies:

The suppliers shall be solely responsible for all taxes, duties, levies, including Octroi, local levies, sales tax, and excise duties incurred as per existing laws in India. No concessional

sales tax form "C" or "D" will be issued by this University. It should be clearly mentioned in the quote whether the prices include taxes & duties. If the price includes taxes and a variation of tax element is sought, then the quote should mention the base level of taxes built into the price.

10. Customs and Excise Duty Exemption:

The University of Hyderabad is under the University Grants Commission (UGC), Government of India, and thus is exempted from excise duty vide GOI notification No.10/97 central excise dated March 1, 1997, and is eligible for concessional customs duty as per GOI notification No.51/96 customs dated 23/07/1996 – amended from time to time.

11. Delivery Earnest Money Deposit (EMD)/Bid Security:

The bidders should furnish bid security for 5% of the estimated cost of the equipment along with their technical bid in the form of Account Payee Demand Draft / Banker's Cheque from any of the Commercial Banks in an acceptable form, drawn in favour of Finance Officer, University of Hyderabad, payable at Hyderabad. If the bidder furnishes a Fixed Deposit Receipt / Bank Guarantee, the instrument shall be issued in favour of Finance Officer, University of Hyderabad, payable at Hyderabad and shall be valid for a period of 135 days from the date of opening technical bids. The MSE units registered with MSME and certificates obtained from NSIC under the Single Point Registration Scheme (SPRS) shall be exempted from payment of Earnest Money Deposit (EMD) on production of requisite proof in respect of valid certification from NSIC.

12. Delivery Location, Delivery Period and Penalty Clauses:

a) The items shall be delivered to: School of Engineering Sciences and Technology University of Hyderabad, Gachibowli, Hyderabad - 500 046, India.

b) In case of foreign suppliers, the firm delivery period (in 12 weeks) after opening an irrevocable Letter of Credit (L.C.) must be given in the Techno-commercial Bid. No part shipment is permitted.

c) In the case of indigenous suppliers, the firm delivery period (in 12 weeks) after the issue of the purchase order must be specified in the technical bid. Delivery is to be made at the University of Hyderabad. Delivery must be on a freight-paid basis for dispatch by road. University of Hyderabad will issue the certificate/declaration instead of a road permit to affect the delivery.

d) In the event of delay in delivery period beyond stipulated period indicated in the Purchase Order, liquidated damages @ 1% per each week of delay subject to a maximum of 5% of the cost of the order will be levied and collected by the University, by way of deduction from the payments due to vendor/supplier.

e) Delay in installation & commissioning beyond the contractual deadline will result in forfeiture of the performance bank guarantee.

13. Warranty:

The supplier must warrant that the goods supplied under the contract are new, unused, and the most recent or current, and incorporate all recent improvements in design and materials as per specifications in this tender document. All vendors (manufacturers/agents) must submit offers with 36 months of comprehensive warranty after installation & successful commissioning of the equipment at the School of Engineering Sciences & Technology, University of Hyderabad, Hyderabad-500046, India, which would be considered for the tender evaluation purpose. The supplier should, however, quote separately for a comprehensive warranty for three successive years after completing a three-year comprehensive warranty. (This is to get an estimate of the cost involved). The warranty offered should be unconditional, and the seller should agree to replace or repair the equipment at his cost. The spare parts for the warranty repair have to be provided by the vendor.

14. Payment :

a) For Indian suppliers and manufacturers - 100% payment subject to submission of 10% performance bank guarantee (PBG) as per clause four will be made direct to Indian suppliers by "Account Payee" cheque against invoice within 30 days of delivery, inspection, successful commissioning & acceptance of equipment/services by the University of Hyderabad.

b) For Foreign suppliers, 100 % by irrevocable foreign letter of credit, subject to the vendor submitting an unconditional PBG from a scheduled bank of 10 % of FOB value, and before opening irrevocable L.C. This PBG must be valid for 60 days after the warranty period.

c) Intermediary agencies, local representatives of foreign manufacturers/suppliers, are involved in importing foreign goods, and an authorization letter/certificate must be obtained to this extent. No agency commission shall be paid to such intermediary agencies/local representatives.

d)In case of L.C.s, bank charges outside India, including confirmation charges, if any, shall be to the supplier's account. The documents for payment shall be presented to the purchaser's bank by the supplier within 21 days of shipment of the equipment.

e)An agreement per standard University norms to be entered with the University before the opening of L.C.

15. Agreement:

An Agreement in the format prescribed by the University of Hyderabad is required to be entered into by the Suppliers or their authorized Representative with the University in respect of purchase of one Bipolar HiPIMS power supply capable of handling reactive sputtering process within a period of 15 days fromthe date of issue of Purchase Order.

16. Risk Purchase Clause:

If the supplier fails to deliver the ordered one Bipolar HiPIMS power supply capable of handling reactive sputtering process within the stipulated delivery period specified in the Purchase Order, the University may resort to procure such items and in such a manner as deemed appropriate, goods or services similar to those undelivered, and the Supplier will be liable to reimburse the University for any excess costs for such similar goods or services.

17. Risk to reject:

The University reserves the right to reject any or all the bids or accept a part of the quotation without assigning any reasons therefor. The decision of the University in this regard is final and binding on all the suppliers.

18. Resolution of disputes:

Any disputes arising out of this contract shall be referred to the University, and if any of the parties here to is dissatisfied with the decision, the dispute shall be referred to the decision of the Arbitrator, who should be acceptable to both the parties, to be appointed by the Vice-Chancellor of the University.The decision of such Arbitrator shall be final and binding on both the parties.

19. Jurisdication of courts:

Disputes, if any, arising during course of execution of the order are to be settled within the jurisdiction of Hyderabad / Ranga Reddy District Courts only.

20. Delay in submission:

The University will not in any way be responsible for any postal / courier delay. Bids received beyond the stipulated date and time of submission are summarily rejected. Tenders incomplete in any respect will be summarily rejected.

21. Opening of tenders/bids:

a) The bids will be opened at the School of Engineering Sciences and Technology, University of Hyderabad, Hyderabad in the presence of bidders or their authorised representatives.

b) The authorised representative shall bring the authorisation letter and submit the same to the University before opening of bids. On failure to do so, the Representative loses the right to participate in the bid opening process.

22. This procurement is entirely governed by the University of Hyderabad Purchase Policy (UHPP). The University of Hyderabad reserves the right to reject any/all quotations or accept any offer or part thereof, without giving any reasons. Its decision in the matter will be final.

23. Any clarifications regarding this tender enquiry should be sought from the undersigned before submission of a formal quotation.

Dr. Venkata Girish Kotnur
School of Engineering Sciences & Technology,
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Central University P.O., Gachibowli,
Hyderabad 500046, INDIA.

Technical Specifications

HiPIMS Power supply capable of handling reactive sputter process.

The proposed Bipolar **High-Power Impulse Magnetron Sputtering (HiPIMS) Power Supply System** shall be a state-of-the-art, digitally controlled pulsed-plasma generation unit designed for advanced thin-film deposition. The system must ensure **high stability, reproducibility, and precision** in both metallic and reactive sputtering environments, in compliance with applicable international electrical and safety standards. All subsystems shall provide full-field, real-time process monitoring to ensure accurate control of plasma characteristics, peak current stability, and power delivery consistency during deposition.

The HiPIMS power supply must be **fully compatible with the existing sputtering system**, including magnetron sources, vacuum chamber interlocks, cooling provisions.. It must integrate seamlessly with existing sputtering facility installed at the University of Hyderabad. The architecture shall be modular and upgradable, enabling future enhancements and addition of advanced features.

Complete package with the following specifications:-

- HiPIMS Power supply bipolar

1.Input Specifications:

- AC Input Voltage: 1-phase + N, 100–240 VAC, 50/60 Hz
- Input Current at 230 V: ≤ 0.5 A
- DC Charging Input: 1000 V max, positive ground
- Trigger Input: 5 V CMOS, ground-referenced
- Communication: RS-232, RS-485, Profibus

2.General output specifications:

- Average power: ≤ 1kW
- Regulation modes: Voltage/Current/Pulse current/Pulse charge
- Frequency range: 50 – 5 kHz or more

- Arc control: < 3μs (reaction time)

3.Negative pulse specifications:

- Peak voltage: 1000V
- Peak current: 100A
- Pulse duration: 4 μs to 1000 μs

4.Positive pulse specifications:

- Peak voltage: ≤ 400 V
- Pulse duration: 2 μs to 500 μs
- Positive pulse delay range: 2 μs to 500 μs

5. Reactive sputter control feature:

- The power supply shall include an integrated process-control mechanism that maintains the reactive HiPIMS discharge at a stable operating point throughout the entire deposition. This method shall function by continuously monitoring the peak current in real time and automatically adjusting the pulse repetition rate to compensate for process drift. The control methodology must be fully embedded within the HiPIMS unit and must not require any external hardware.
- For real-time discharge monitoring, the power supply should have BNC outputs for both discharge voltage and discharge current.

6. Negative Output DC Unit Specifications:

- DC Voltage: -1000V
- Nominal power: upto 1.5 kW
- Current max.: upto 1.5 A
- Remote mode: 0-10V (external)

7. Positive Output DC Unit Specifications:

- DC Voltage: upto 300V
- Nominal power: upto 200 W
- The unit shall support bidirectional power flow and include regenerative load capability

Note:

1. Materials needs to handled: Metals, nitrides and oxides
2. Regulatory compliance: CE & CSA marked and conform to electromagnetic compatability.

Installation and Training at no extra cost

The system should be installed and commissioned by the supplier at School of Engineering Scieences and Technology, University of Hyderabad, Gachibowli, Hyderabad-500046, Telangana and give hands-on training and trial tests on the use of the system.

Signature of Principal Investigator





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